Light

GCSE Computer Science - Cyber Security Threats (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of cyber security?

- ○ **A.** To make computers run faster
- ○ **B.** To protect systems and data from attack, damage, and unauthorised access
- ○ **C.** To increase the storage capacity of hard drives
- ○ **D.** To improve the quality of internet connections

Q2 (1 mark)

Why is the human element often considered the weakest link in cyber security?

- ○ **A.** Humans type too slowly on keyboards
- ○ **B.** Humans require too much electricity to work
- ○ **C.** Humans can be tricked or manipulated into bypassing security measures
- ○ **D.** Humans cannot see computer screens clearly

Q3 (1 mark)

How do phishing attacks typically try to create urgency?

- ○ **A.** By offering slow delivery of products
- ○ **B.** By providing detailed technical documentation
- ○ **C.** By suggesting the victim take their time to respond
- ○ **D.** By threatening negative consequences like account closure or missed deliveries if the victim does not act immediately

Q4 (1 mark)

What preparation typically distinguishes blagging from other social engineering attacks?

- ○ **A.** The attacker researches the target to create a believable, personalised scenario
- ○ **B.** Blagging requires no preparation at all
- ○ **C.** Blagging only works on computers built before 2010

- ○ **D.** The attacker must be in the same country as the victim

Q5 (1 mark)

In which situations might shouldering (shoulder surfing) commonly occur?

- ○ **A.** Only in swimming pools
- ○ **B.** Only when using desktop computers at home alone
- ○ **C.** Only during thunderstorms
- ○ **D.** At cash machines, on public transport, in coffee shops, or anywhere screens and keypads are visible to others

Q6 (1 mark)

What action causes a computer virus to become active and start spreading?

- ○ **A.** The user running or opening the infected file or program
- ○ **B.** Turning the computer off
- ○ **C.** Connecting a printer to the computer
- ○ **D.** Changing the desktop wallpaper

Q7 (1 mark)

Why is trojan malware named after the Trojan Horse from Greek mythology?

- ○ **A.** Because it only affects computers in Greece
- ○ **B.** Because it makes computers run as fast as horses
- ○ **C.** Because like the wooden horse that concealed Greek soldiers, it appears harmless but hides a malicious purpose inside
- ○ **D.** Because it was invented by someone named Troy

Q8 (1 mark)

Why does spyware typically try to remain hidden from the user?

- ○ **A.** Because if detected, the user would remove it, stopping the attacker's surveillance
- ○ **B.** Because hidden software runs faster
- ○ **C.** Because visible software uses more electricity
- ○ **D.** Because computer law requires all software to be invisible

Q9 (1 mark)

Why is pharming potentially more dangerous than standard phishing?

- ○ **A.** Pharming only works on older computers
- ○ **B.** Pharming affects fewer people than phishing
- ○ **C.** Pharming attacks are always announced in advance

○ **D.** The victim types the correct website address but is still redirected to a fake site, making detection much harder

Q10 (1 mark)

What might attackers gain access to by exploiting weak passwords?

○ **A.** Email accounts, bank accounts, social media profiles, company systems, and personal files

○ **B.** Nothing of value, since passwords only protect unimportant data

○ **C.** Only the computer's screensaver settings

○ **D.** The ability to make the computer's keyboard louder

Q11 (1 mark)

What types of devices are included when discussing 'removable media' as a cyber security threat?

○ **A.** Only floppy disks from the 1990s

○ **B.** USB flash drives, external hard drives, memory cards, CDs, DVDs, and any portable storage device

○ **C.** Only devices that are painted red

○ **D.** Only devices manufactured by one specific company

Q12 (1 mark)

Why do attackers specifically target systems running unpatched software?

- ○ **A.** Unpatched software has a more attractive user interface
- ○ **B.** Unpatched systems contain known vulnerabilities with published exploitation methods
- ○ **C.** Patched software cannot connect to the internet
- ○ **D.** Attackers prefer older software because it runs faster

Q13 (2 marks)

Tom receives a text message saying: 'ROYAL MAIL: Your parcel could not be delivered. Pay the £1.50 redelivery fee here: bit.ly/rm-pay2day'. Tom is not expecting any parcels. What type of attack is this most likely to be?

- ○ **A.** Pharming
- ○ **B.** A legitimate delivery notification from Royal Mail
- ○ **C.** Phishing via SMS (sometimes called smishing), attempting to steal payment details
- ○ **D.** Shouldering

Q14 (2 marks)

What is the key difference between a virus and spyware in terms of their primary purpose?

- ○ **A.** A virus is designed to replicate and spread to other files and systems, while spyware is designed to secretly collect and transmit information about the user
- ○ **B.** There is no difference; they are the same type of malware
- ○ **C.** Spyware replicates more quickly than viruses

○ **D.** Viruses only affect mobile phones while spyware only affects computers

Q15 (2 marks)

Why must penetration testing only be performed by authorised professionals with explicit permission?

○ **A.** Because penetration testing equipment is very expensive

○ **B.** Because without authorisation, attempting to access systems using hacking techniques is illegal, regardless of intent

○ **C.** Because only professionals know how to use keyboards

○ **D.** Because penetration testing only works during business hours

Q16 (2 marks)

How do attackers exploit default passwords to compromise large numbers of devices quickly?

○ **A.** They use automated scripts that scan the internet and try known default credentials for common device types

○ **B.** They visit each device in person and try passwords manually

○ **C.** They send polite emails asking device owners to share their passwords

○ **D.** They wait for device owners to publish their passwords on social media

Q17 (2 marks)

A manager asks their assistant to send some emails on their behalf, so they share their login password. Why is this practice a security risk?

- ○ **A.** The computer will automatically delete all emails when a password is shared
- ○ **B.** The assistant might type the password incorrectly
- ○ **C.** Sharing passwords makes the computer run more slowly
- ○ **D.** Password sharing eliminates individual accountability, expands the attack surface if either person is compromised, and violates security policies

Q18 (2 marks)

A company allows employees to bring personal USB drives from home and connect them to work computers. What security risks does this create?

- ○ **A.** No risks, since personal USB drives are always safe
- ○ **B.** Personal drives might carry malware from home computers, and employees could use them to steal company data
- ○ **C.** USB drives can only store music files, so there is no risk
- ○ **D.** Personal USB drives make computers display more colours

Q19 (2 marks)

Which password demonstrates the strongest security practices?

- ○ **A.** password123
- ○ **B.** john1985 (using the user's name and birth year)

- ○ **C.** qwerty
- ○ **D.** K9$mPx2!vLnQ (random combination of uppercase, lowercase, numbers, and symbols)

Q20 (2 marks)

Why do phishing emails often impersonate well-known brands like Amazon, PayPal, or major banks?

- ○ **A.** Because these companies have given permission for their brands to be used in phishing
- ○ **B.** Because smaller companies have better security
- ○ **C.** Because almost everyone has accounts with at least one major brand, increasing the chance recipients will believe the message is legitimate and relevant to them
- ○ **D.** Because major brands send their emails in a special secret format

Q21 (2 marks)

What malicious actions might a trojan perform once installed on a victim's computer?

- ○ **A.** Create a backdoor for remote access, download additional malware, steal data, record keystrokes, or join the computer to a botnet
- ○ **B.** Only improve the computer's performance as promised
- ○ **C.** Only change the desktop wallpaper to a picture of a horse
- ○ **D.** Only send helpful tips to the user about computer maintenance

Q22 (2 marks)

Why are regular software updates essential for cyber security?

- ○ **A.** Updates only add new features that users want
- ○ **B.** Updates make software more colourful
- ○ **C.** Updates are only necessary for games, not security software
- ○ **D.** Updates fix known security vulnerabilities before attackers can exploit them, closing gaps that could allow system compromise

Q23 (2 marks)

Which psychological principles do social engineers commonly exploit?

- ○ **A.** Trust, authority, fear, urgency, curiosity, and the natural human desire to be helpful
- ○ **B.** Only the victim's knowledge of advanced mathematics
- ○ **C.** Only the victim's preference for certain colours
- ○ **D.** Only the victim's ability to type quickly

Q24 (3 marks)

A school's website is defaced with offensive content, and the attackers have also downloaded the student database containing names, addresses, dates of birth, and medical information. Analyse the full scope of this security incident.

- ○ **A.** This is only a minor inconvenience with no serious consequences

○ **B.** The website defacement damages school reputation, but the data breach is far more serious: children's personal data is compromised, exposing minors to identity theft; medical information disclosure violates data protection laws; parents will lose trust; regulatory fines are likely; and the stolen data could be misused for years

○ **C.** Only the website appearance has been affected

○ **D.** Schools cannot be targets of cyber attacks

Q25 (3 marks)

A financial services company needs to assess two different risks: (1) whether criminals could breach their public-facing website from the internet, and (2) whether a rogue employee with basic system access could access restricted customer financial data. Which penetration testing approaches should they use for each scenario?

○ **A.** Both scenarios require testing with full administrator credentials

○ **B.** Neither scenario requires penetration testing; antivirus software is sufficient

○ **C.** For the website breach risk: testing with no prior credentials (external attack simulation); For the rogue employee risk: testing with basic employee credentials (insider threat simulation)

○ **D.** Only the website needs testing; insider threats cannot be tested

Q26 (3 marks)

A finance department employee receives an urgent email appearing to be from the company CEO: 'I'm in a confidential meeting and cannot take calls. I need you to process an immediate wire transfer of £47,000 to a new supplier. This is time-sensitive-complete it within the hour and confirm by reply. Do not discuss with anyone else due to confidentiality.'

The CEO is genuinely travelling that day. Identify the threats and why this attack might succeed.

- ○ **A.** This is a routine legitimate business request
- ○ **B.** Only shouldering is involved in this attack
- ○ **C.** This is CEO fraud (a form of blagging/phishing) exploiting authority, urgency, secrecy, and real knowledge of the CEO's schedule; it might succeed because the employee fears disobeying an executive, the urgency prevents verification, and the confidentiality instruction isolates the victim
- ○ **D.** This attack would never work because employees always verify wire transfers

Q27 (3 marks)

A retail shop has these security issues: (1) The point-of-sale system runs Windows 7, which Microsoft stopped supporting in 2020; (2) The staff WiFi uses the password 'shopwifi123' which is written on a sticky note at the counter; (3) Employees have not received any training on recognising suspicious emails. Analyse the combined risks these vulnerabilities create.

- ○ **A.** These are all best practices that improve security
- ○ **B.** The unsupported Windows 7 system processing payments has known vulnerabilities that could allow card data theft; the weak, visible WiFi password allows anyone to access the network and potentially reach the payment system; untrained staff might fall for phishing that installs malware-together these create a path from simple password observation to customer payment card theft
- ○ **C.** Only the WiFi password is a minor concern
- ○ **D.** Retail shops cannot be affected by cyber attacks

Q28 (3 marks)

A charity discovers that a trojan has been present on their donor management system for six months. Investigation reveals the trojan contained spyware that captured and transmitted donor names, addresses, email addresses, and payment card details to external servers. The charity has 15,000 donors on their database. Analyse the potential impacts of this attack.

- ○ **A.** The only impact is that donors might receive some extra emails
- ○ **B.** This is a minor technical issue with no real consequences
- ○ **C.** Severe impacts include: 15,000 donors face identity theft and payment card fraud; the charity must notify all affected donors and regulators under data protection law; donors will lose trust and may stop giving; regulatory fines are likely for inadequate data protection; the charity faces potential legal action from affected donors; reputation damage may affect future fundraising permanently; and attackers may have ongoing access if the trojan created a backdoor
- ○ **D.** Charities cannot be targets of cyber attacks

Q29 (3 marks)

A company is implementing remote working. They propose: (1) employees use personal laptops without company security software; (2) work files are shared via personal email accounts; (3) employees can connect from any public WiFi. Evaluate these security implications.

- ○ **A.** These are excellent security practices for remote work
- ○ **B.** All three proposals create severe security risks: personal laptops may lack protection and carry existing malware; personal email bypasses company security and data protection controls; public WiFi allows traffic interception-together they create an environment where company data is effectively unprotected and exposed to multiple attack vectors
- ○ **C.** Only the WiFi policy is slightly concerning

○ **D.** Remote working cannot affect security

Q30 (3 marks)

A customer service representative receives a phone call from someone claiming to be from the company's IT department: 'We've detected unusual activity on your account. I need to verify your identity and run some security checks. First, please confirm your username and password so I can access your system remotely.' The caller sounds professional and knows the representative's name and department. Identify all the threats and explain why verification is essential.

○ **A.** This is standard IT support procedure and should be followed

○ **B.** This is vishing (voice phishing/phone-based blagging) using authority impersonation and false urgency; providing credentials would give attackers access to company systems; the knowledge of name and department likely came from reconnaissance; IT departments never ask for passwords-verification through official channels is essential because internal phone numbers can be spoofed

○ **C.** Phone calls cannot be used for social engineering

○ **D.** If the caller sounds professional, they must be legitimate IT staff

Total: **55** marks • Time: **30 min**

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